SIMATS ENGINEERING

ASSESSMENT TOOL 2 — FILLED ANSWERS

Course: Artificial Intelligence (CSA1708) | CO1

Q1. Hospital Shift Assignment — Backtracking Search

Variables: D1, D2, D3 (one per doctor).

Domain: Morning (M), Afternoon (A), Night (N), with M < A < N.

Constraints: D1 ≠ N; shift(D2) < shift(D3); all shifts different; D3 ≠ M.

Search order: D1 → D2 → D3; values M → A → N.

Backtracking trace:

Step	Assignment	Result
1	D1 = M	Accept
2	D1=M, D2=M	Reject — M already used
3	D1=M, D2=A	Accept
4	D1=M, D2=A, D3=M	Reject — D3 cannot be M; duplicate
5	D1=M, D2=A, D3=A	Reject — duplicate; A is not before A
6	D1=M, D2=A, D3=N	Solution found
7–8	D1=M, D2=N	No valid D3; backtrack
9–10	D1=A, D2=M, D3=N	Second solution
11	D1=A, D2=N	No valid D3; backtrack
12	D1=N	Reject — D1 cannot be N

Final valid schedules: (D1=M, D2=A, D3=N) and (D1=A, D2=M, D3=N). Backtracking returns the first one encountered.

First solution: D1 = Morning, D2 = Afternoon, D3 = Night.

Q2. Robot in a 5×5 Grid — BFS and A*

Grid assumptions: S=(0,0), G=(4,4). Obstacles: (0,3), (1,1), (1,3), (2,1), (3,3). Moves are Up, Down, Left, Right; each move costs 1.

Manhattan heuristic: h(n)=|row–4|+|col–4|; h(S)=8. It is admissible and consistent.

Part A — BFS:

BFS expands nodes in FIFO order. The resulting shortest path is:

(0,0) → (1,0) → (2,0) → (3,0) → (4,0) → (4,1) → (4,2) → (4,3) → (4,4)

Cost = 8 moves. BFS is optimal because every move has equal cost.

Part B — Priority Queue with Manhattan Distance (A*):

Use f(n)=g(n)+h(n), where g is path cost so far and h is Manhattan distance.

A* path: (0,0) → (0,1) → (0,2) → (1,2) → (2,2) → (2,3) → (2,4) → (3,4) → (4,4)

Cost = 8 moves. Since the optimal cost equals h(S)=8, the path is optimal. BFS and A* give different shortest paths, but both have cost 8.

Q3. Autonomous Rescue Robot — Partially Observable Online Search

a) Constraints and effects on decision-making

Constraint	Effect
Up, Down, Left, Right	Four-action grid; branching factor ≤ 4; moves are reversible.
Obstacles	Blocked cells are illegal; the robot must discover them and re-plan.
Adjacent-cell sensing only	Environment is partially observable; the robot maintains a map/belief and decides with incomplete information
Move cost 1; risky cell costs 3	Weighted-cost problem; UCS/A* is appropriate. A longer safe route may be cheaper than entering a risky cell
Minimise total cost while staying safe	Avoid known dangerous cells unless the cost saving justifies the risk; never enter known blocked cells.
Online updating	Sense → update map → plan → act → re-plan when needed.

b) Solution approach — Online Uniform Cost Search (UCS) with re-planning

1. Initialise the map with S and G; unknown cells are initially treated as free with cost 1.
2. Sense the four adjacent cells and label each as free, obstacle, or risky.
3. Run UCS on the current knowledge map. Normal cells cost 1; risky cells cost 3; obstacles are removed.
4. Execute only the first move of the cheapest plan.
5. Sense again. If new information invalidates or increases the cost of the plan, discard it and re-plan.
6. Stop when G is reached. If no safe path exists, explore the unknown frontier or report failure.
7. Store visited cells/map information to avoid repeating dead ends.

Optional improvement: A* with Manhattan distance can reduce node expansions while remaining optimal.

c) Application of AI concepts

The robot is a goal-based and utility-based intelligent agent following the sense → think → act loop.

PEAS element	Description
Performance measure	Reach the survivor at G with minimum total path cost and no unsafe moves.
Environment	Disaster-affected building represented as a grid with obstacles and risky zones.
Actuators	Movement motors: up, down, left, right.
Sensors	Short-range sensors that observe adjacent cells only.

Rationality: The robot selects the cheapest safe action based on its current percept history and knowledge, avoiding known obstacles and risky cells unless justified by cost.

Environment type: Partially observable; single-agent; deterministic for known cells; sequential; mostly static; discrete; unknown.

d) Why online UCS is suitable

• Handles the +2 risk penalty, unlike BFS which treats all steps equally.

• On the known map, UCS is optimal and complete because all step costs are positive.

• Re-planning supports partial observability and adjacent-cell sensing.

• Risky cells naturally become more expensive, encouraging safer routes.

• Newly discovered obstacles can be added to the map and the cheapest path recomputed.

• For larger grids, A* or D* Lite with Manhattan heuristic can improve efficiency.

Conclusion: Online UCS with dynamic map updates provides a suitable lowest-cost, safety-aware approach for a partially observable rescue environment.